



Edge Deployment in Robotics

A comprehensive guide to implementing AI at the edge for autonomous robotic systems

Robots are fundamentally mobile edge computing systems that must make critical decisions locally with limited resources. This presentation explores how to effectively deploy AI on robotics platforms where reliability, latency, and efficiency are paramount.

Why Edge Deployment?

Cloud-based processing introduces unacceptable latency for real-time robotic control systems that require immediate response.

→ Offline Capability

Robots must maintain functionality in environments with poor or non-existent connectivity

→ Safety & Autonomy

Local inference ensures critical safety functions remain operational regardless of network status

→ Cloud-Edge Synergy

Cloud supports long-term learning and fleet-wide intelligence, not real-time control loops



Core Requirements of Edge AI in Robotics

Low-Latency Inference

Safety-critical control loops require processing speeds of $<10\text{-}50\text{ms}$ to ensure responsive operation

Examples: Collision avoidance, dynamic balancing, gripper force control

Energy Efficiency

Battery-powered robots demand optimized computation to maximize operational runtime

Power budget directly impacts robot size, weight, and mission duration

Resilience

Must maintain core functionality when network connectivity is degraded or unavailable

Graceful performance degradation rather than catastrophic failure

Security

Local data processing often involves sensitive information about environments and operations

Physical access to robots creates unique security challenges

Typical Edge Hardware



NVIDIA Jetson Series

Nano, Xavier, and Orin platforms offering GPU-accelerated robotics computation

15-50W power envelope, ideal for medium-sized robots

Intel Movidius / OpenVINO

Specialized vision acceleration units optimized for lower power applications

Qualcomm RB5

Integrated 5G connectivity with robotics-focused system-on-chip

Custom ASICs

Application-specific chips like Tesla FSD and DJI flight controllers

Edge Deployment Workflow



Train Models

Utilize high-performance cloud GPU clusters for initial model training with large datasets



Optimize Models

Apply quantization, pruning, and distillation techniques to reduce model size and computational requirements



Package Format

Convert to deployable formats like ONNX, TensorRT, or TFLite compatible with edge hardware



OTA Deployment

Securely deploy packaged models to robot fleets with version control



Local Inference

Execute models on robot with safety-critical watchdog monitoring

Model Optimization for Edge

Techniques for Maximizing Performance

Quantization

Reduce numerical precision from FP32 $\rightarrow$ FP16 $\rightarrow$ INT8

Up to 4x memory reduction with minimal accuracy loss

Pruning

Systematically remove redundant or low-importance neurons

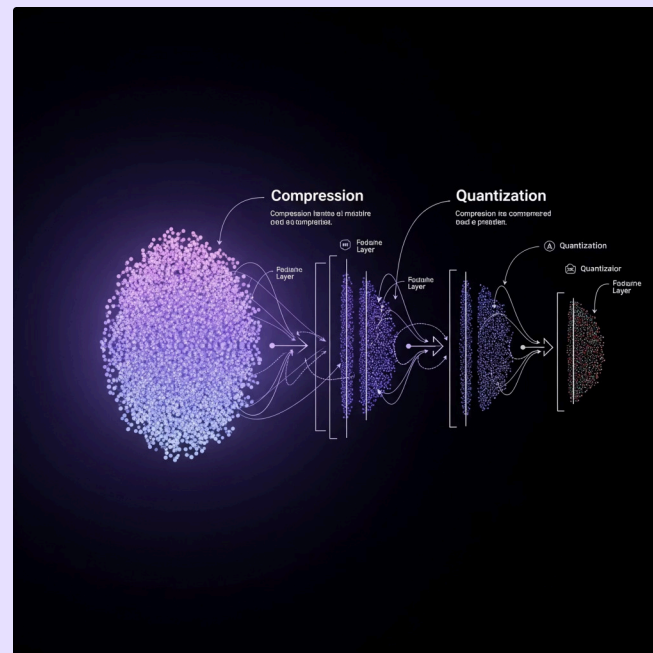
Can reduce model size by 50-90% for many vision tasks

Knowledge Distillation

Train compact "student" models that mimic larger "teacher" networks

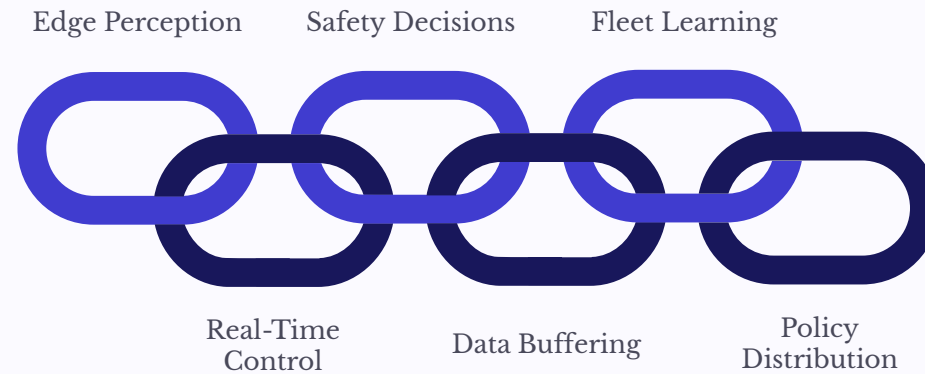
Runtime Acceleration

Leverage specialized libraries: TensorRT, TFLite, ONNX Runtime



Optimization typically trades a small decrease in accuracy for significant improvements in inference speed and energy efficiency

Communication Balance: Edge vs. Cloud



The ideal robotics architecture creates a **symbiotic relationship** between edge and cloud:

Edge Responsibilities

- Real-time perception and control
- Safety-critical decision making
- Short-term data buffering
- Immediate environment interaction

Cloud Responsibilities

- Large-scale model training
- Fleet-wide learning and policy updates
- Simulation testing and validation
- Long-term data storage and analysis

Middleware for Edge Deployment

Specialized software infrastructure connects AI models with robotic hardware and enables efficient operation at the edge



ROS/ROS2

Robot Operating System provides a flexible framework for orchestrating nodes, services, and messages across robotic subsystems

Enables modular deployment of AI components



DDS

Data Distribution Service provides real-time, reliable messaging for mission-critical applications

Underpins ROS2's communication layer



KubeEdge

Extends Kubernetes to edge devices, enabling container orchestration and consistent deployment across robot fleets



MQTT

Lightweight publish/subscribe messaging protocol optimized for high-latency or unreliable networks

Ideal for robot-cloud telemetry

Case Study: Warehouse Robots



System Architecture

- **Edge Hardware:** NVIDIA Jetson Orin runs perception, planning, and navigation stacks
- **Compute Distribution:** Object detection and path planning run entirely on-robot
- **Edge Resilience:** Robots maintain full functionality during network outages
- **Cloud Integration:** Central simulation environment tests navigation improvements before deployment
- **Synchronization:** Fleet-wide OTA updates deploy improved navigation models during charging cycles

✓ **Result:** 99.8% operational availability with 24/7 capability regardless of network status

Case Study: Agricultural Drones

Precision Agriculture at the Edge

Modern agricultural drones require sophisticated on-board AI to analyze crop health and make real-time treatment decisions, even in remote fields with no connectivity.

Edge Processing

Onboard vision system uses TFLite-optimized models for real-time crop health analysis

INT8 quantization enables operation on low-power CPUs with 12W power envelope

Cloud Training

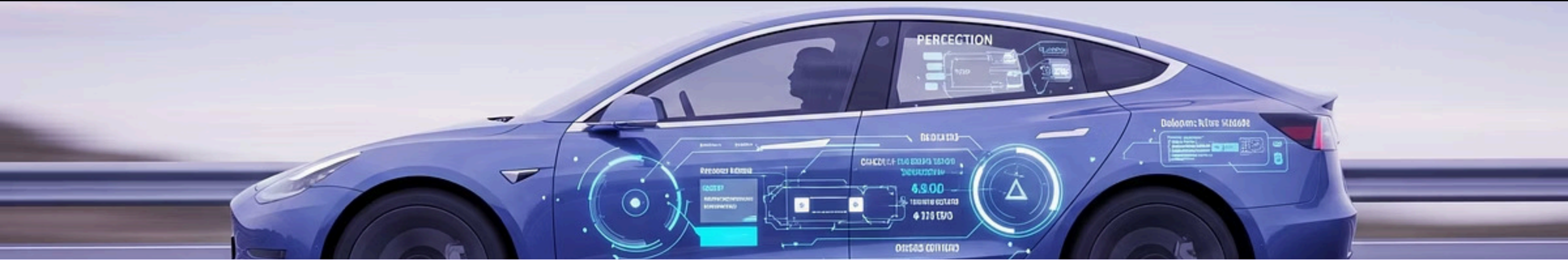
Models trained on massive datasets combining satellite imagery, ground truth samples, and historical yield data

Real-Time Decision Making

Edge AI controls variable-rate application of fertilizer and pesticides based on plant conditions



Challenges: Limited payload capacity restricts computing hardware, requiring extreme optimization



Case Study: Tesla Full Self-Driving

Edge Implementation

- Custom FSD chip designed specifically for neural network inference
- 144 TOPS of compute in ~75W power envelope
- Runs complete vision and planning stack locally
- Dedicated redundant processing paths for safety-critical functions

Local inference ensures that safety-critical operations like emergency braking and lane keeping function regardless of connectivity status.

Cloud-Edge Loop

- Dojo supercomputer trains massive neural networks
- Vehicles upload edge cases and driver interventions
- Models optimized specifically for custom silicon
- OTA updates deliver incremental improvements

Security in Edge Robotics

Edge deployment creates unique security challenges as robots operate in physical environments and may be accessible to potential attackers.

Data Protection

Implement local data encryption to protect sensitive sensor data, environment maps, and operational parameters

Secure Updates

OTA model updates must be cryptographically signed and verified to prevent malicious model injection

Network Security

Zero-trust networking principles for robot-to-cloud communication with mutual TLS authentication

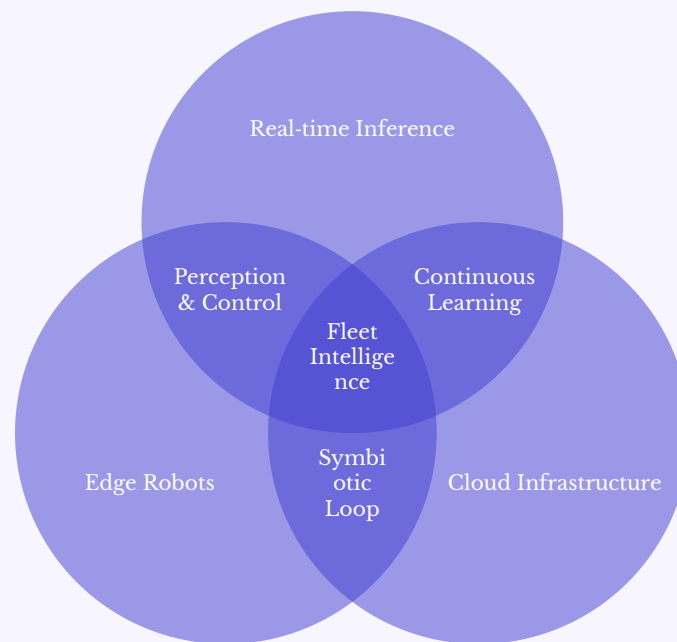
Physical Security

Tamper detection mechanisms to prevent unauthorized hardware modifications or sensor spoofing



 Security vulnerabilities in robotic systems can lead to both data breaches and physical safety hazards

Robots Think Locally, Learn Globally



Edge inference runs in milliseconds, cloud provides continuous learning

The ideal architecture creates a harmonious loop: robots make independent decisions locally while contributing to collective intelligence globally.



Challenges in Edge Deployment



Compute Trade-offs

Limited computational resources force difficult trade-offs between model accuracy and inference latency

Every millisecond matters in safety-critical applications



Energy Constraints

High-performance AI accelerators like GPUs consume significant power

Battery technology advances lag behind computing needs



Deployment Complexity

OTA updates must be managed carefully to avoid operational downtime

Fallback mechanisms essential for production systems



Sim2Real Gap

Models trained in simulation often perform poorly in real-world conditions

Domain adaptation techniques required for robust performance

Summary

Edge deployment in robotics is fundamentally about creating **autonomous, safe, and resilient** systems that can function in the real world—not just in controlled laboratory environments.

Success requires a holistic approach:

- Carefully optimized models matched to efficient hardware
- Thoughtful balance between edge and cloud responsibilities
- Robust deployment pipelines with comprehensive testing
- Security considerations at every level

The future belongs to robotic systems that combine the immediacy of edge inference with the collective intelligence of cloud learning.



Questions?